Sort — Merge Sort

Merge sub-process

Start: two sorted sub-arrays, left half and right half

Create pointers i = 0 (start of left half), j = 0 (start of right half), k = left (write position in the original array)

Have both i reached the end of the left half AND j reached the end of the right half?

No

Has i NOT reached the end of the left half, AND (j has reached the end of the right half OR leftHalf[i] is less than or equal to rightHalf[j])?

Yes

Copy leftHalf[i] into array[k]; increment i and k

No

Copy rightHalf[j] into array[k]; increment j and k

Yes

Merge complete

Start: unsorted array, with a left index and right index defining the range to sort

Is left greater than or equal to right?

No

Calculate mid = (left + right) divided by 2

Recursively call Merge Sort on the left half (left to mid)

Recursively call Merge Sort on the right half (mid+1 to right)

Merge the two now-sorted halves back together

Yes

Return — a single element (or empty range) is already sorted

invoke merge